

Wi-Fi Series

Data Sheet

USE_8733

Wi-Fi / BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	06/05/2026	MB	NN	Initial release

Contents

Features.....	3
General.....	3
MCU Platform.....	3
WLAN Features.....	3
Host Interface.....	3
Bluetooth Features.....	3
General Description:.....	4
Pinout Diagram:.....	4
Pin Descriptions:.....	4
Electrical Specifications.....	6
Power Supply DC Characteristics:.....	6
Power Consumption.....	6
RF Characteristics.....	6
1. 2.4G RF Specification.....	6
2. 5GHz RF Specification.....	7
5GHz(20MHz) Channel table.....	8
3. Bluetooth Specification.....	9
Block Diagram.....	10
Reference Design.....	11
Physical Dimensions.....	12
Layout Recommendation.....	13
Module Dimension:.....	13
Recommended Reflow Profile :.....	14

Features :

General

- Support Wi-Fi/Bluetooth functionalities
- IEEE 802.11a/b/g/n compliant
- Bluetooth 5.2 compliant
- 1T1R architecture
- USB interface for WLAN/BT
- Dimension: 12.2 × 12.9 × 2.37 mm
- Operating temperature: 0°C to 70°C
- Storage temperature: -40°C to 85°C

MCU Platform

- Chipset: RTL8733BU-CG

WLAN Features

- USB Multi-Function for both BT (USB function 0) and WLAN (USB function 1)

Host Interface

- USB2.0 interface for WLAN
- USB2.0 interface for Bluetooth

Bluetooth Features

- Supports Bluetooth multiple Low Energy states
- Bluetooth 5.2 Dual Mode support: Simultaneous LE and BR/EDR
- Small SMT package

General Description:

The USE_8733 is a highly integrated module with Realtek RTL8733BU-CG single-chip 802.11a/b/g/n 1T1R WLAN and integrated Bluetooth 5.2 combo chip with USB 2.0 multi-function support.

The highly integrated module enables web browsing, VoIP, and video streaming applications. With seamless roaming capabilities and advanced security features, the module can interact with different vendors' 802.11a/b/g/n 1x1 Access Points in wireless LAN environments.

The wireless module complies with IEEE 802.11a/b/g/n standards and supports data rates up to 150Mbps when operating with 40MHz bandwidth. The integrated module provides USB 2.0 interface support for Wi-Fi functionality.

For more information, contact us at info@unisemindia.com.

Pinout Diagram:

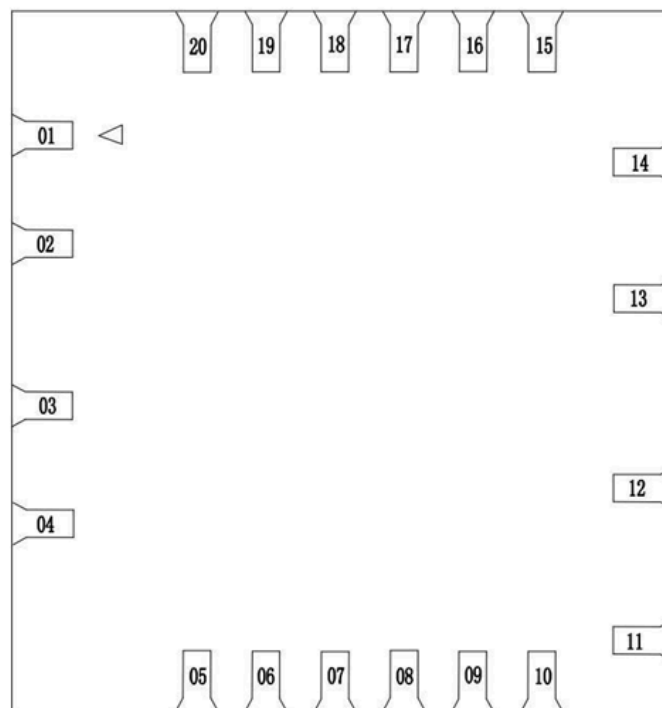


Fig1: Module Pinouts

Pin Descriptions:

Pin No	Pin Name	Type	Description
1	GND	—	Ground connections

2	RF_0	I/O	RF I/O, Wi-Fi 2.4G/5G/BT
3	NC	—	Floating (NC)
4	GND	—	Ground connections
5	BT_PCM_IN	I	PCM data input
6	BT_PCM_OUT	O	PCM data output
7	BT_PCM_SYNC	I/O	PCM sync signal
8	BT_PCM_CLK	I/O	PCM clock
9	BT_WAKE_HST	O	Bluetooth device wake-up HOST
10	HST_WAKE_BT	I	HOST wake-up Bluetooth device
11	VD33	P	Main power source input
12	DM	I/O	USB2.0 differential pair D-
13	DP	I/O	USB2.0 differential pair D+
14	GND	—	Ground connections
15	NC	—	Floating (NC)
16	NC	—	Floating (NC)
17	NC	—	Floating (NC)
18	CHIP_EN	I	Enable pin for module. ON: pull high; OFF: pull low (Default: pull high)
19	HST_WAKE_WL	I	HOST wake-up WL device
20	WL_WAKE_HST	O	WL device wake-up HOST

The signal type codes are used in the table:

- P:POWER
- I:INPUT
- O:OUTPUT

Electrical Specifications :

Power Supply DC Characteristics:

Parameter	MIN	TYP	MAX	Unit
Operating Temperature	0	25	70	°C
VDD33	3.0	3.3	3.6	V

Power Consumption

Power Consumption	TX HT40 11n Mode/14dbm	102.4mA
	TX HT20 11n Mode/14dbm	113.6mA
	TX HT20 11g Mode/15dbm	115.6mA
	TX HT20 11b Mode/17dbm	216.8mA
	RX Mode	80mA

RF Characteristics

1. 2.4G RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11 b/g/n Wi-Fi compliant	
Frequency Range	2.400 GHz ~ 2.4835 GHz (2.4 GHz ISM Band)	
Number of Channels	2.4GHz : Ch1 ~ Ch14	
Test Items	Typical Value	EVM
Output Power	802.11b /11Mbps : 17dBm ± 2 dB	EVM £ -9dB
	802.11g /54Mbps : 15dBm ± 2 dB	EVM £ -25dB
	802.11n /MCS7 : 14dBm ± 2 dB	EVM £ -28dB
Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	±20ppm	
Test Items	TYP Test Value	Standard Value
SISO Receive Sensitivity (11b,20MHz) @8% PER	-11Mbps PER @ -85 dBm	≤-81

SISO Receive Sensitivity (11g,20MHz) @10% PER	-54Mbps PER @ -71 dBm	≤-67
SISO Receive Sensitivity (11n,20MHz) @10% PER	-MCS=7 PER @ -68 dBm	≤-65
SISO Receive Sensitivity (11n,40MHz) @10% PER	-MCS=7, PER @ -65 dBm	≤-62
Maximum Input Level	802.11b : -10 dBm	
	802.11g/n : -20 dBm	

Note: Other data rates transmit power are controlled by the Power-by-Rate function of the driver.

2. 5GHz RF Specification

Feature	Description	
WLAN Standard	IEEE 802.11 a/n Wi-Fi compliant	
Frequency Range	5.150 GHz ~ 5.850 GHz (5.0 GHz Band)	
Number of Channels	5.0GHz : Please see the following table1	
Test Items	Typical Value	EVM
Output Power	802.11a /54Mbps : 15dBm ± 2 dB	EVM ≤ -25dB
	802.11n /MCS7 : 14dBm ± 2 dB	EVM ≤ -28dB
Spectrum Mask	Meet with IEEE standard	
Freq. Tolerance	±20ppm	
Test Items	TYP Test Value	Standard Value
SISO Receive Sensitivity (11a,20MHz) @10% PER	-54Mbps PER @ -71 dBm	≤-68
SISO Receive Sensitivity (11n,20MHz) @10% PER	-MCS=7 PER @ -68 dBm	≤-65

SISO Receive Sensitivity (11n,40MHz) @10% PER	-MCS=7, PER @ -65 dBm	≤-62
Maximum Input Level	802.11a/n : -10 dBm	
Antenna Reference	Small antennas with 0~2 dBi peak gain	

Note: Other data rates transmit power are controlled by the Power-by-Rate function of the driver.

5GHz(20MHz) Channel table

Band range	Operating Channel Numbers	Channel center frequencies(MHz)
5150MHz~5250MHz	36	5180
	40	5200
	44	5220
	48	5240
5250MHz~5350MHz	52	5260
	56	5280
	60	5300
	64	5320
5470MHz~5725MHz	100	5500
	104	5520
	108	5540
	112	5560
	116	5580
	120	5600
	124	5620
	128	5640
	132	5660
	136	5680
	140	5700
	149	5745
	153	5765

5725MHz~5850MHz	157	5785
	161	5805
	165	5825

3. Bluetooth Specification

Feature	Description
General Specification	
Bluetooth Standard	BDR(1Mbps),EDR(2Mbps & 3Mbps),LE(1Mbps),2LE(2Mbps)
Host Interface	UART
Frequency Band	2400 MHz ~ 2483.5 MHz
Number of Channels	79 channels for classic,40 channels for BLE
Modulation	GFSK, $\pi/4$ -DQPSK,8DPSK
RF Specification	
Output Power , tolerance ± 3 dB	
	CL1(dBm)
BDR Output Power	5
EDR Output Power	5
BLE Output Power	5
Sensitivity, tolerance : /	
Sensitivity @ BER=0.1% for GFSK (1Mbps)	-92
Sensitivity @ BER=0.01% for $\pi/4$ -DQPSK (2Mbps)	-86
Sensitivity @ BER=0.01% for 8DPSK (3Mbps)	-85
Sensitivity @ BLE=30.8% for LE (1Mbps)	-90

Sensitivity @ BLE=30.8% for 2LE (2Mbps)	-90
Maximum Input Level	GFSK (1Mbps):-20dBm
	$\pi/4$ -DQPSK (2Mbps) :-20dBm
	8DPSK (3Mbps) :-20dBm

Block Diagram :

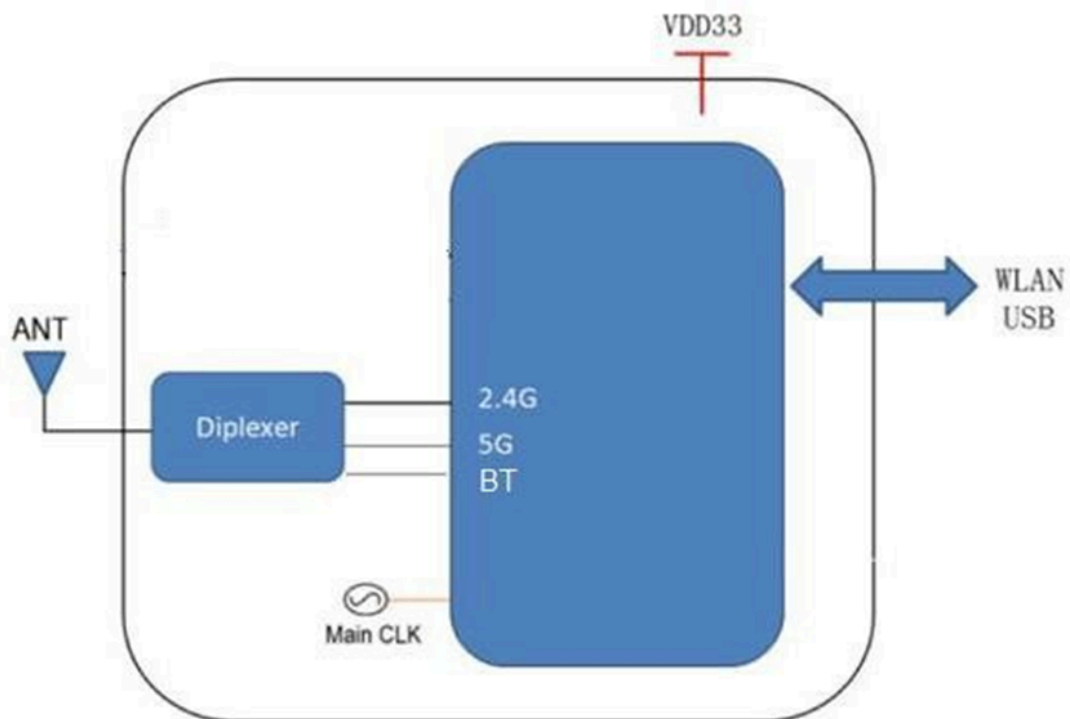
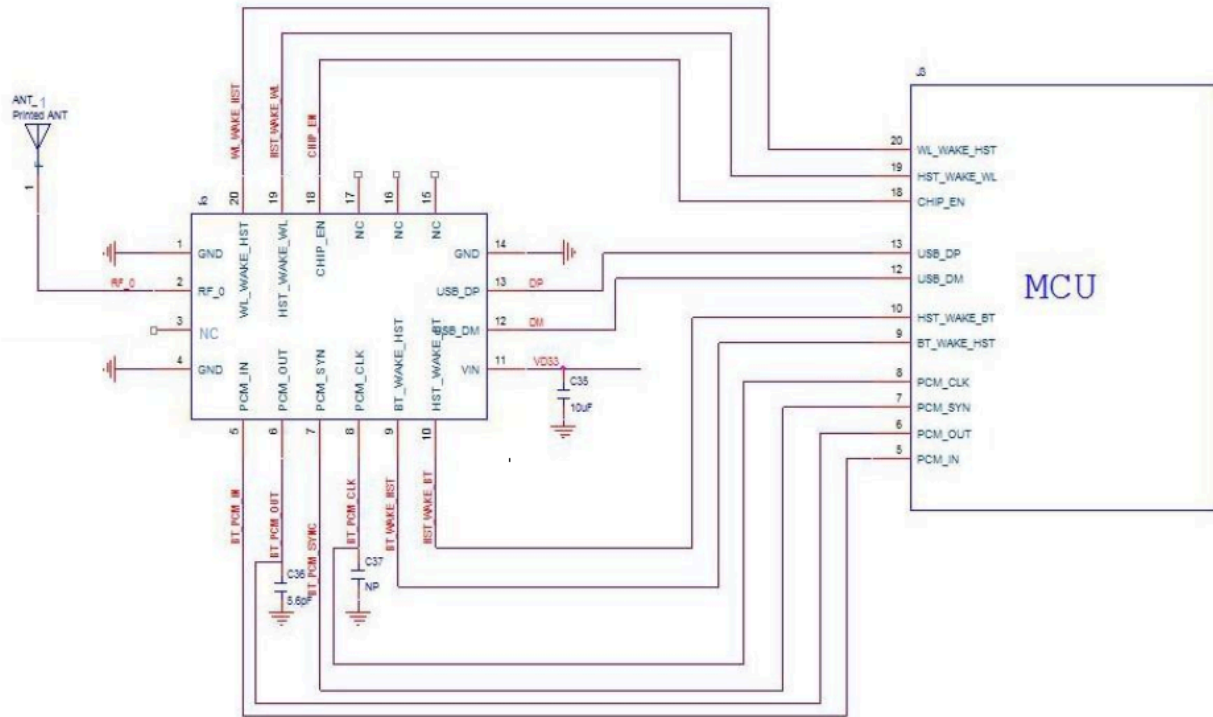


Fig 2 : Block Diagram

Reference Design :



Note: Module requires independent power supply , supply capacity $\geq 1200\text{mA}$ and ripple less than 100mV ; Do not share power with amplifier, infrared device, camera, etc. USB differential trace, please keep 90 ohm.

Physical Dimensions :

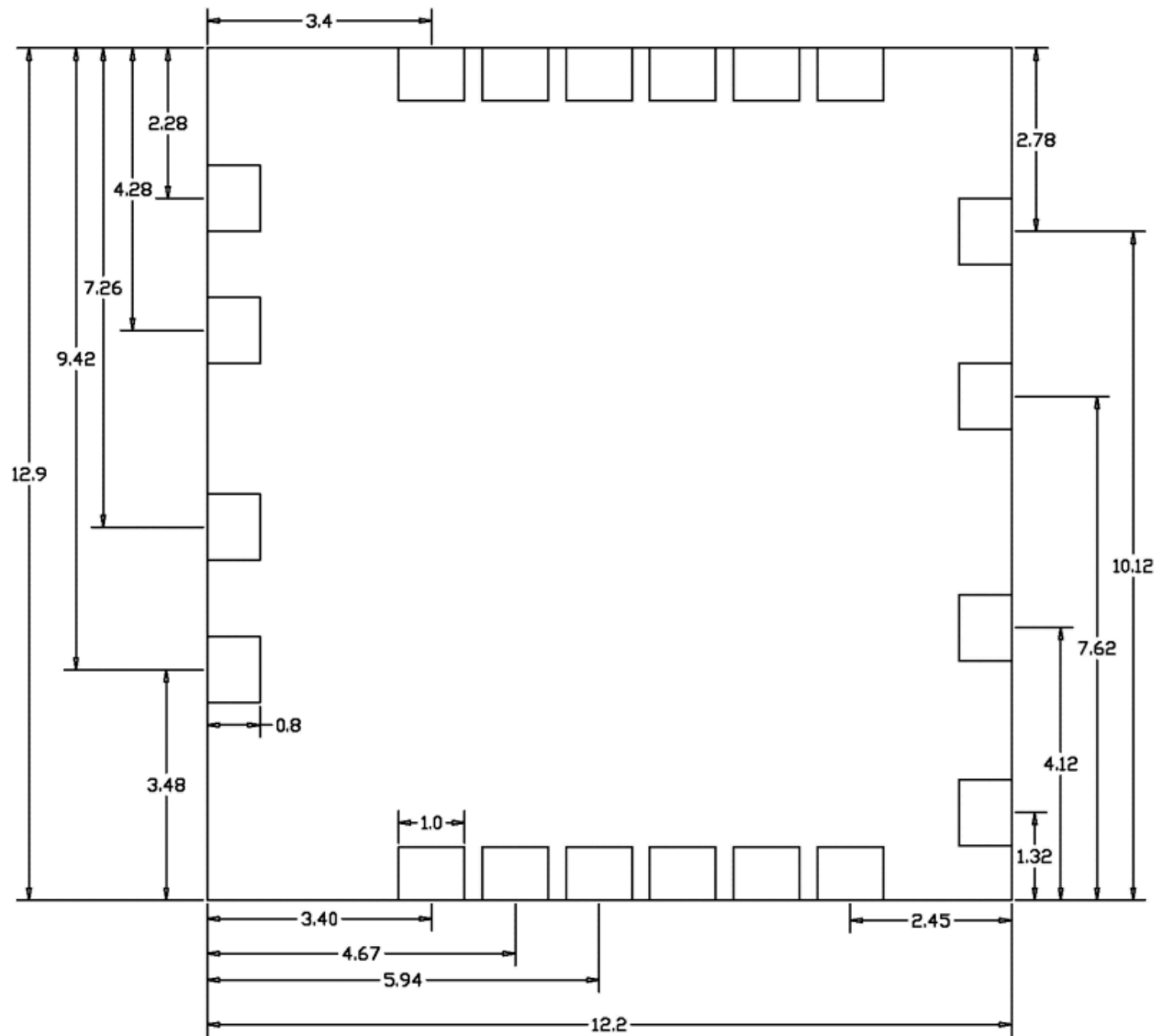


Fig 3 : Physical Diagram (top view)

Layout Recommendation:

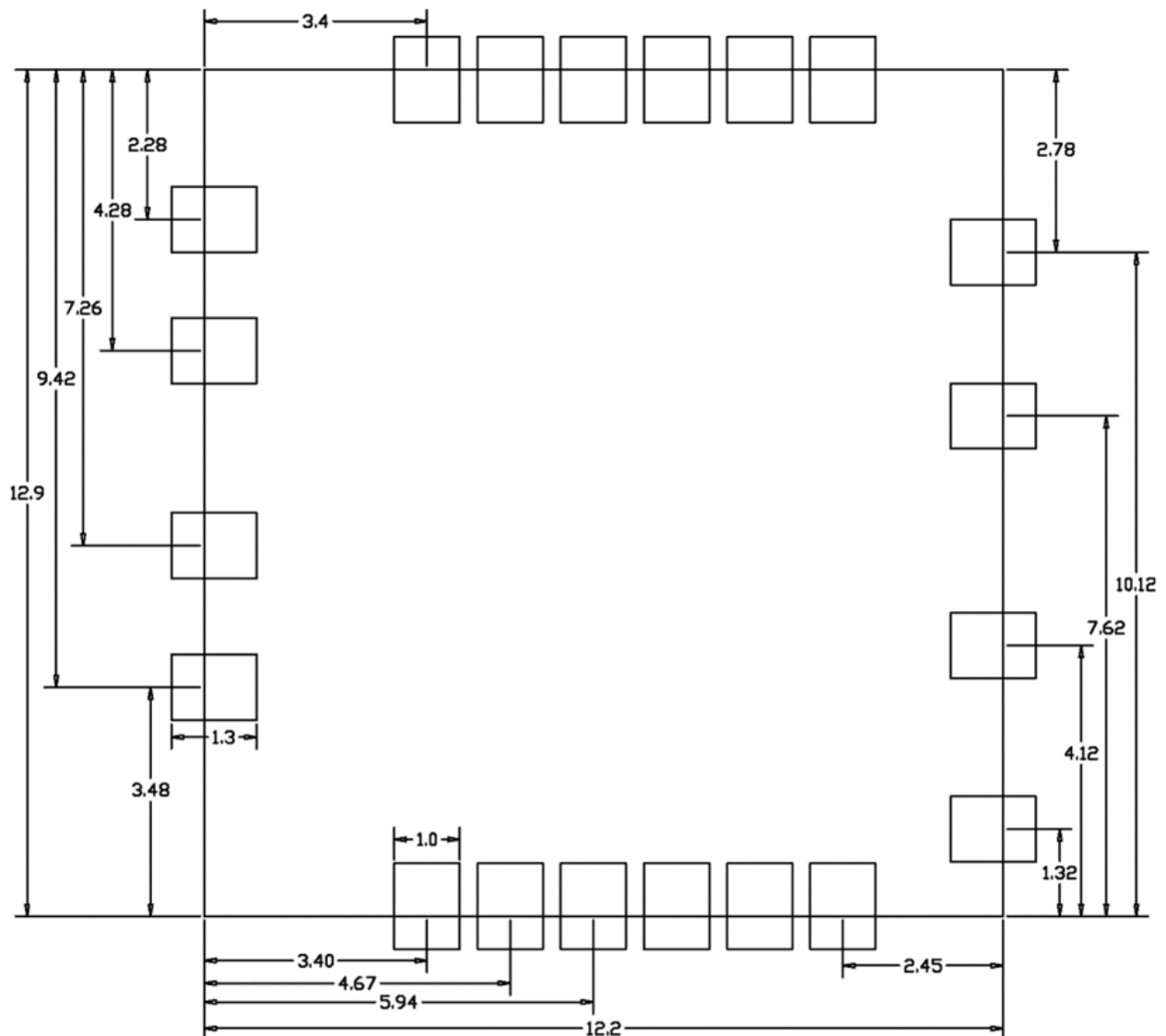


Fig 4 : Layout Diagram

Module Dimension:

- L x W : 12.2 x 12.9 (+0.3/-0.1) mm

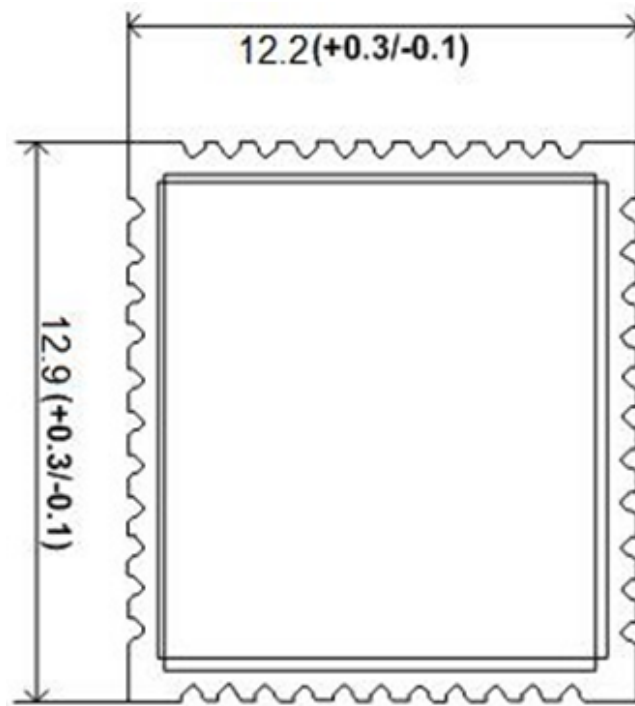


Fig 5(a) : L x W

- H: 2.37 (± 0.2) mm



Fig 5(b) : Height

Recommended Reflow Profile :

Referred to IPC/JEDEC standard

Peak Temperature : $< 260^{\circ}\text{C}$

Number of Times : ≥ 2 times

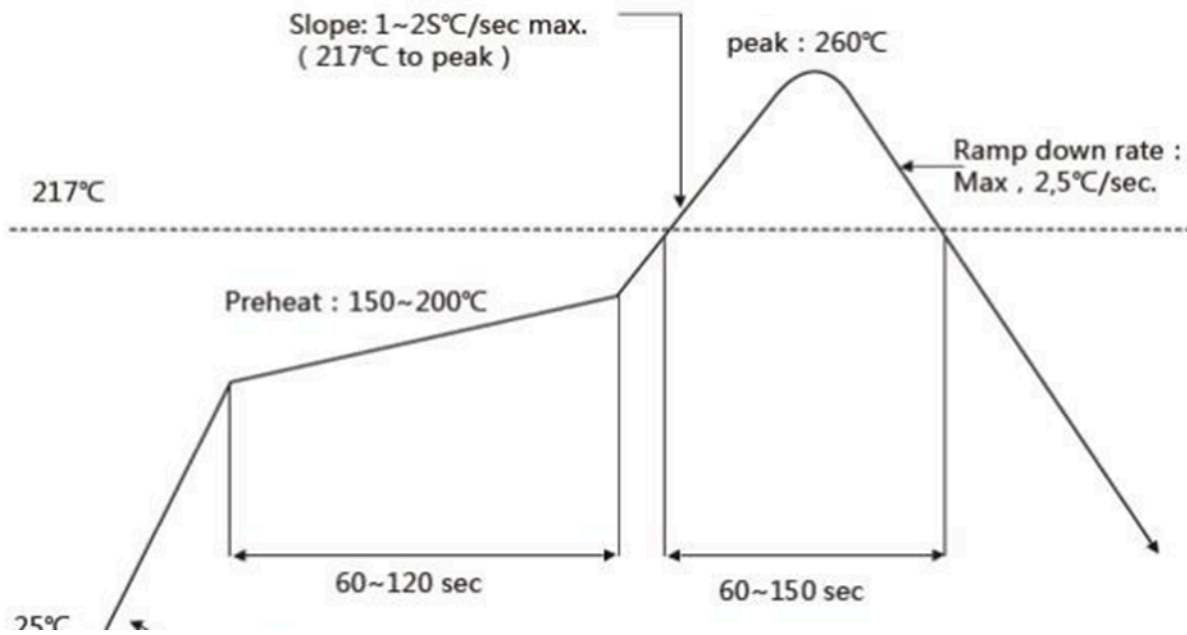
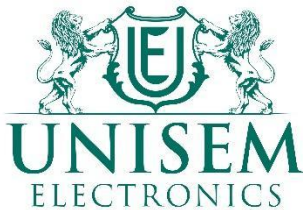


Fig 6 : Recommended Reflow Profile

For ordering USE_8733 please contact us on info@unisemindia.com.



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